ROADWATCH

INVENTION DISCLOSURE - TECHNICAL DETAIL REPORT

Prepared from the current Roadwatch source code and publicly accessible background materials. Complete inventor/ownership details and institutional IP review before filing.

Important completion notes

- Inventorship and ownership must be confirmed by the institutional IP/technology-transfer office. The listed student contact is provisional and does not settle legal inventorship.
- The current software is a browser-side mapping and maintenance-status interface. It does not include an embedded trained vision model, live camera/GPS ingestion, or durable server storage.
- A preliminary public search located closely related papers and patents. This document does not claim patentability, freedom to operate, or legal novelty.
- Before filing, replace bracketed fields with confirmed information and attach labeled/unlabeled drawings as required by the institute.

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1. Title of the Invention (15 words maximum)

Roadwatch: A Geospatial Dashboard for Municipal Road Defect Prioritization

2. What is the Invention?

Roadwatch is a browser-based software system for organizing road-defect records on a geographical map and presenting them through a linked table, summary indicators, filters, and a maintenance-status workflow. The problem addressed is that road-condition reports can be difficult to compare and route for follow-up when location, severity, time, and repair status are spread across separate views. The current solution consolidates these fields into one operator interface. The current source is a client-side application; it does not itself perform road-image inference or acquire live GPS data.

3. Search Terms

Road damage monitoring; pothole management dashboard; geospatial road maintenance; GPS-tagged road defects; municipal asset management; severity-based work-order prioritization; Leaflet mapping; road anomaly reporting; computer-vision road inspection; pavement condition monitoring.

4a. Background of the Invention - Present State of Art

Existing approaches include manual road inspections, citizen or agency reporting applications, vehicle-mounted image capture, accelerometer/GPS-based sensing, computer-vision models, GIS road-asset systems, and municipal maintenance work-order tools. Published research has reported detection and classification of road damage from smartphone images [1], while the RDD2022 dataset supports research across six countries, including India [2]. Commercial and public-sector applications can combine reporting, GPS, mapping, and maintenance notification. Prior patents also describe location-tagged pothole characteristics, severity, mapping, and repair reporting [8], [9].

4b. Limitations of Existing Technologies

Manual reporting can be slow and inconsistent. Vision systems depend on training coverage, image quality, location accuracy, and environmental robustness. Separate reporting, GIS, and work-order tools can create duplicated records and weak traceability. Some approaches require specialized sensing equipment, data plans, or back-end integration. The current Roadwatch software itself has important limitations: the record list is configured in the HTML, state changes are not persisted to a server, and no detector is embedded.

5. Prior Art

Relevant prior art includes Maeda et al. (2018), who describe deep-neural-network road damage detection from smartphone images [1]; RDD2022, a multi-national image dataset for road-damage detection [2]; U.S. Patent 9,416,499 B2, “System and method for sensing and managing pothole location and pothole characteristics” [8]; and U.S. Patent 9,365,217 B2, “Mobile pothole detection system and method” [9]. These sources overlap with the broad concepts of detection, geolocation, characterization, mapping, and maintenance reporting. They must be reviewed by an IP professional before any claim of novelty or inventiveness.

6. Has the Work Filled a Major Gap in Prior Art?

A potential product-integration gap is the need for a simple municipal operator interface that presents defect locations, common filtering, workload summaries, and a status-transition path in one view. The current work implements this interface integration. It has not been established that this is a patentable gap; existing systems and patent documents may cover similar combinations. A structured claim chart and formal novelty search are necessary.

7. Components of the Invention

(1) Browser user interface written in HTML/CSS; (2) client-side road-defect record collection; (3) state and filter functions for time range, search, severity, status, and sorting; (4) Leaflet map layer and CARTO basemap; (5) severity-coded circle markers and pop-ups; (6) KPI, severity and detection-volume summaries; (7) linked detection table; (8) detail drawer; (9) status-action functions

for scheduling, repair, dismissal, reopening, and restoration; and (10) configured crew labels. Planned, not yet integrated, components include a detection API, validated GPS source, database, authentication, event history, and notifications.

8. How Does the Invention Work?

The page initializes the theme and the Leaflet map, then loads the locally configured defect records. Each record carries an identifier, location labels, coordinates, severity, confidence, time, status, crew, depth, and width fields. A user selects a time range; the application scopes the records and applies the active status/severity chip and search query. The same visible collection is used to render the map markers and detection table. Selecting a marker or row opens the detail drawer. A user can invoke a status action. The client-side object is updated, associated KPIs and views are re-rendered, and a visible notification reports the action. In the current implementation these updates remain in browser memory; they do not automatically synchronize to a persistent database or trigger an actual crew dispatch.

9. Novelty / Inventive Step

Potential differentiating aspects to investigate are the specific interaction design that connects map selection, tabular filtering, severity summaries, and record-level maintenance-state transitions in a single lightweight interface. These components rely on known web mapping and dashboard techniques. No assertion is made that the combination is new or non-obvious. Patentability requires a claim-specific prior-art review and counsel guidance.

10. Advantages and Improvements

The interface consolidates defect fields in one place, reduces switching between a map and a table, makes high-severity records easier to inspect, provides selectable reporting windows, exposes a record-level status path, and uses standard browser technologies. Leaflet and the selected basemap ecosystem reduce the need to build map rendering from scratch. Expected operational advantages such as shorter repair times, reduced costs, or improved safety have not been measured and should not be stated as results until evaluated.

11. Alternative Forms / Preferred Embodiments

Possible alternatives include a mobile-first inspector interface; a municipal API-backed web application; integration with a vehicle-mounted camera and GNSS receiver; ingestion from an existing road-survey database; a dashboard for multiple cities; configurable severity schemes; heat maps and road-segment aggregation; duplicate clustering by proximity and time; offline collection with synchronization; and role-based work-order management. These are possible future embodiments, not current capabilities.

12. Drawings Required

Attach Fig. 1 (system architecture), Fig. 2 (record-processing and operator workflow), Fig. 3 (screen layout showing map, severity summary, KPIs, table, and drawer), and Fig. 4 (record state lifecycle). Each drawing should be provided in two versions: one labeled and numbered, and one without labels. The architecture and workflow illustrations included in this report are conceptual and should be checked against the implementation before filing.

13. Brief Description of Drawings

Fig. 1 shows a defect-record source feeding shared filters and state logic, which feed the map, table, and KPI presentation; maintenance actions operate on records and server persistence is a planned extension. Fig. 2 describes creation or receipt of a record, filtering, visualization, triage, and subsequent persistent storage/notification as a planned step. Fig. 3 should identify the map, legend, filter chips, KPI row, detection table, and record drawer in the interface. Fig. 4 should show the status transitions new/in review to scheduled, then repaired, with dismissal and restoration/reopening branches.

14. Complete Description and Working Examples

A browser loads the Roadwatch HTML file and its Leaflet stylesheet/script. After initialization, it creates a Leaflet map centered on Chennai and a layer group for records. A record is rendered as a circle marker whose radius corresponds to severity. Repaired records are rendered with outline treatment. Pop-ups identify the record and summarize its street, area, severity, status, and confidence. The page calculates KPI totals and severity counts from the current date-filtered collection. A table search matches road, area, or identifier; selectable chips constrain records by work status or severity. Sorting operates on severity rank, confidence, or detection time. The detail drawer presents location, time, dimensions, and crew assignment, with available actions depending on current status. Example state transitions include new to scheduled, scheduled to repaired, and repaired to review after reopening. These actions are local client updates. A deployed version should validate API inputs and use transactions, authorization, timestamps, and an audit log.

15. Experiments / Third-Party Validation / Equations

No controlled road survey, independent detector benchmark, third-party validation, or field repair trial has been provided. The current evidence is limited to the corrected interactive map loading and displaying markers during user-side verification. This does not validate the physical presence, exact location, severity, or dimensions of every road defect. No detector accuracy, precision, recall, F1 score, coordinate error, latency, uptime, or repair-cost saving is claimed. For a future detector evaluation, report precision = TP/(TP+FP), recall = TP/(TP+FN), and F1 = 2PR/(P+R), with TP/FP/FN defined at an agreed IoU or location-matching threshold. For location validation, report geodesic distance between predicted and surveyed coordinates and summarize median and 95th-percentile error.

16. Instructions / Code / Algorithm

Code type: HTML5, CSS3, and vanilla JavaScript. Map library: Leaflet 1.9.4 loaded from a CDN. Basemap: CARTO Voyager raster tiles with OpenStreetMap attribution. The current application is organized within one HTML file. Key functions include record construction (`mk`), time scoping (`inRange`), chip matching (`matchChip`), text matching (`matchSearch`), shared list creation

(`scoped`/`visible`), map initialization and rendering (`initMap`/`renderMap`), KPI rendering (`renderKPI`), table rendering (`renderTable`), detail drawer (`openDrawer`), and status transitions (`act`). No neural-network code, weights, training script, backend endpoint, or database schema is integrated in the current source.

17. Training Data and Machine-Learning Model

No ML model is currently integrated into Roadwatch, and no project-specific training run is claimed. Therefore, there is no current model architecture, trained weights, training split, or model accuracy to report. A future vision subsystem could evaluate RDD2022, which contains road images and labeled damage instances from six countries including India [2], alongside locally collected and reviewed Chennai road images. Data licensing, annotation quality, class balance, location coverage, and domain shift must be reviewed. Training should use separated train/validation/test sets, location-aware splits to reduce leakage, and metrics such as precision, recall, F1, and inference latency.

18. Public Disclosure / Publication

To be confirmed by all inventors. List any public presentations, university submissions, posters, Git repositories, social-media posts, publications, external exhibitions, releases, or disclosures made before filing, including dates and URLs. Do not assume that a private exchange or source-code sharing establishes public disclosure; the institutional IP office should advise on the relevant facts and deadlines.

19. Stage / Level of Development

Current status: a working browser-side application with interactive map tiles, record markers, filters, summary indicators, a table, a detail drawer, and client-side status actions. Not completed: live camera inference, device GPS ingestion, authenticated API, persistent server database, system-level testing, third-party validation, and field deployment. Provisional readiness assessment: TRL 3 (proof-of-concept level). The institution should confirm the applicable TRL using its formal criteria.

20. Aspects Proposed for Protection

For discussion with the IP office only: (a) a computer-implemented municipal road-defect review interface that binds geographical markers and structured records to common time/status/severity filters and workload summaries; (b) a record-selection workflow that links map or table selection to maintenance-state actions; and (c) future variants where verified defect evidence, location provenance, duplicate resolution, and audited repair-state updates are integrated. These draft concepts are not legal claims and may be unpatentable or covered by prior art. Do not file without counsel review.

21. Preliminary Novelty / Inventiveness Search

A preliminary public scan was conducted on Oct. 11, 2026, using public literature pages and Google Patents. It is not a comprehensive patent search and not a freedom-to-operate opinion.

No.	Reference	Existing idea described	Potential point to investigate for Roadwatch
1	Maeda et al., 2018 [1]	Deep-learning detection/classification from smartphone road images.	Roadwatch is an operator dashboard; no detection network currently integrated.
2	RDD2022, 2024 [2]	Multi-country road-damage data for training and evaluation.	Potential future data source; does not establish novelty of the dashboard.
3	US9416499B2 [8]	Sensing/managing pothole location, characteristics, severity and repair reporting.	Substantial overlap in the broad monitoring workflow; claim analysis is essential.
4	US9365217B2 [9]	Mobile pothole detection with location and road mapping.	Location-based road-defect monitoring is established prior art.
5	US11829959B1	Autonomous pothole detection, mapping and repair decisions.	Further review required for any proposed claim around automated prioritization.

22. Would a Person of Average Skill Have Arrived at the Invention?

This cannot be answered confidently without claim-level patent analysis. Because related systems already use GPS-tagged defect records, maps, severity information, and repair workflows, a broad claim to those functions may face substantial prior art. Any protectable contribution would need to be narrowly defined and supported by a non-obvious technical effect established with the institution IP office.

23. Broad Workable Parameter Ranges

Current code-level ranges and values: geographic validation bounds in the helper function are latitude 12.7–13.3 and longitude 79.9–80.5; map initial center approximately 13.05 N, 80.22 E and initial zoom 11; tile layer max zoom 20; supported time filters 24 hours, 7 days, and 30 days; confidence fields represented from 0 to 1; severity classes minor/moderate/severe/critical; marker radii approximately 6–10 pixels; configured depth fields 2–13 cm and width fields 20–96 cm. The depth/width figures are the current record values, not a validated measurement envelope. These should not be treated as engineering tolerances or field-verified parameters.

24a. Potential Commercialization Organizations

The following are potential organizations to approach because their published activities relate to mapping, engineering, industrial technology, or infrastructure. This listing does not mean any organization has expressed interest.

Organization	Address	Potential relevance
MapmyIndia / C.E. Info Systems Ltd.	237, Okhla Industrial Estate, Phase III, New Delhi 110020, India.	Digital maps, navigation and location-based services.
L&T Technology Services Ltd.	L&T Knowledge City, SEZ, West Block II, NH 8, Village Ankhol, Vadodara, Gujarat 390019, India.	Engineering and digital systems integration.
Tata Elxsi Ltd.	ITPB Road, Whitefield, Bengaluru 560048, India.	Software, design and digital engineering.
Siemens Ltd.	Birla Aurora, Level 21, Plot No. 1080, Dr. Annie Besant Road, Worli, Mumbai 400030, India.	Infrastructure and smart-city technology.
Bosch Ltd.	P.O. Box 3000, Hosur Road, Adugodi, Bengaluru 560030, India.	Sensors, mobility and engineering systems.

24b. Short Marketing Profile

Roadwatch is designed as an operator-facing road-defect dashboard for municipal road maintenance teams. It combines a map, severity view, filtered detection table, concise workload indicators, and record-level status actions. Its intended value is straightforward review and triage without building a custom map engine. A marketable release would need validated field locations, secure APIs, persistent storage, integration with an approved detection/reporting source, operational support, and municipal pilots. The current implementation does not establish quantitative savings or commercial readiness.

25. Market Potential, Adoption and Growth

Relevant adoption drivers include continued investment in urban infrastructure and smart-city services [7], road-safety priorities [6], improved availability of vehicle-mounted cameras and GNSS, and growing use of computer vision for road inspection [1]–[3]. A defensible monetary market-size estimate specific to Roadwatch has not been established. The table below is a qualitative market-planning scaffold, not a market forecast.

Year	Estimated market size	Growth drivers	Notable existing approaches
2026	Not quantified for Roadwatch; requires market research.	Municipal digitization, urban-road maintenance, GIS adoption.	Manual inspection, mobile reporting, GPS-tagged survey systems.
2027	To be estimated from verified procurement and pilot data.	Camera-based inspection, open mapping ecosystems, work-order integration.	Road-damage datasets/models, survey vehicles, GIS dashboards.
2028	To be validated; do not infer from general market reports alone.	Integration, data quality, lifecycle tracking, multi-city deployment.	Computer-vision platforms and municipal asset-management systems.

26. Inventors Involved

Sidak Singh [provisional; confirm the contribution of every team member and any co-inventors with the institute IP office before filing]. Use the legal inventorship standard, not project membership alone.

27a. Potential Users

Municipal road and public-works departments, road-maintenance coordinators, field inspection teams, civil engineers, infrastructure analysts, approved contractors, and (in a future extension) public reporting teams.

27b. User Age Group

The professional interface is intended for trained adult users. The underlying technology is not age-specific. A public-facing reporting channel, if created, would require accessibility, privacy, and safety review.

27c. Expected Benefits

Centralized visibility of road-defect records, easier identification of high-severity items, consistent time/status filtering, improved traceability of review decisions, and a potential foundation for maintenance planning. These are expected benefits and have not been measured in a field trial.

27d. Cost Advantage

The browser interface uses widely available web technologies and Leaflet, which lowers the need to develop a proprietary map-rendering engine. Costs remain for hosting, tile-provider usage, secure backend services, device/sensor integration, data labeling, maintenance, and operational support. No total-cost comparison has been conducted.

28. Technology Readiness Level (TRL)

Provisional selection: TRL 3 - proof-of-concept level. The browser-side interface has been implemented and opened successfully, but it has not been validated in a field-relevant environment with verified field coordinates, a live detection feed, or persistent maintenance records. Confirm the rating with the institution.

29. Additional Notes or Remarks

Before filing: confirm all inventors and ownership; document funding and institutional resources; list all public disclosures with dates; run a professional novelty and freedom-to-operate search; verify the coordinate provenance; provide source-code version/hash; attach labeled and unlabeled drawings; collect systematic browser test evidence; and consult the institution IP office before further public disclosure. This report is a technical draft, not a legal opinion.

Annex A. System Architecture

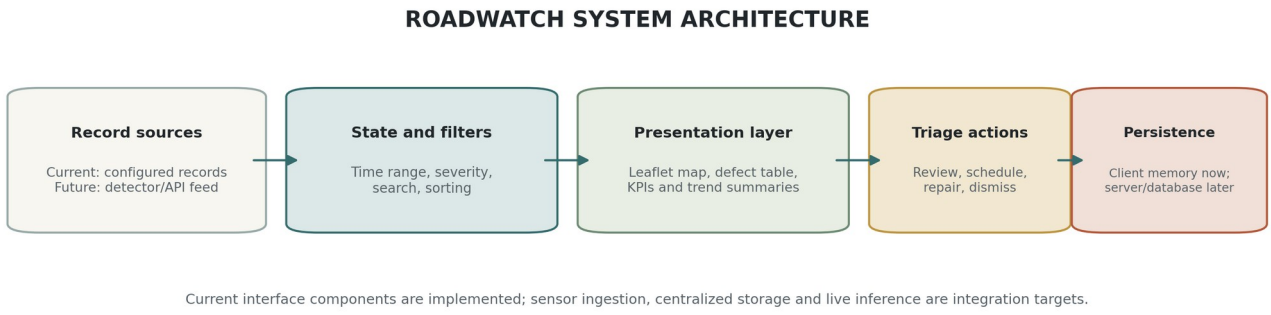


Fig. A1. Architecture showing current browser-side components and future service integrations.

Annex B. Workflow

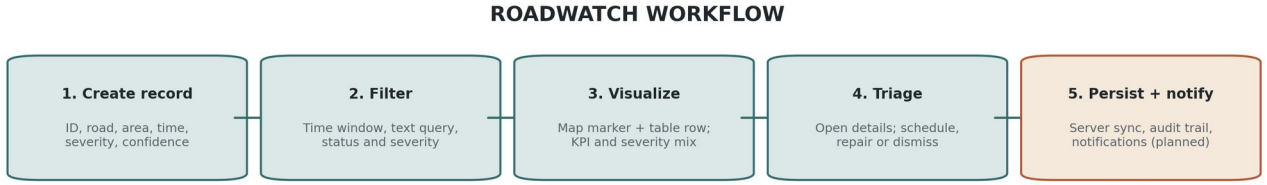


Fig. B1. Workflow from a defect record to filtering, visualization and triage.

Sources Used for Preliminary Technical Context

[1] H. Maeda, Y. Sekimoto, T. Seto, T. Kashiyama, and H. Omata, “Road damage detection and classification using deep neural networks with smartphone images,” Computer-Aided Civil and Infrastructure Engineering, vol. 33, no. 12, pp. 1127–1141, 2018, doi: 10.1111/mice.12387.

[2] D. Arya et al., “RDD2022: A multi-national image dataset for automatic road damage detection,” Geoscience Data Journal, vol. 11, no. 4, pp. 846–862, 2024, doi: 10.1002/gdj3.260.

[3] S. H. Abdelwahed et al., “Advancements in real-time road damage detection: a comprehensive survey of methodologies and datasets,” Journal of Real-Time Image Processing, vol. 22, Art. no. 137, 2025, doi: 10.1007/s11554-025-01683-1.

[4] Leaflet contributors, “Leaflet Quick Start Guide,” Leaflet documentation. [Online]. Available: https://leafletjs.com/examples/quick-start/ (accessed Oct. 11, 2026).

[5] OpenStreetMap contributors, “Copyright and License.” [Online]. Available: https://www.openstreetmap.org/copyright (accessed Oct. 11, 2026).

[6] United Nations, “Goal 11: Make cities and human settlements inclusive, safe, resilient and sustainable.” [Online]. Available: https://sdgs.un.org/goals/goal11 (accessed Oct. 11, 2026).

[7] Ministry of Housing and Urban Affairs, Government of India, “Smart Cities Mission Achievements,” Press Information Bureau, Dec. 18, 2024. [Online]. Available: https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=2085711.

[8] “System and method for sensing and managing pothole location and pothole characteristics,” U.S. Patent 9,416,499 B2, Aug. 16, 2016. [Online]. Available: https://patents.google.com/patent/US9416499B2/en.

[9] “Mobile pothole detection system and method,” U.S. Patent 9,365,217 B2, Jun. 14, 2016. [Online]. Available: https://patents.google.com/patent/US9365217B2/en.